



Kinetic studies in pyrolysis of garden waste in the context of downdraft gasification: Experiments and modeling

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ABSTRACT

The disposal and management of dry leaves litter (biomass) from roads and gardens in urban areas is a challenging task. It is beneficial to use this waste as an energy source in thermal applications through pyrolysis and gasification. Downdraft gasifier is one such promising reactor that can be employed to achieve this task in a decentralized manner. A distinct pyrolysis zone is established in the gasifier during its operation, and it influences the performance significantly. Hence, the present work is undertaken to study the kinetics of pyrolysis in conditions that prevail in gasifier. A fixed-bed laboratory reactor that uses biomass pellets is employed to analyze the gases generated during pyrolysis process. Carbon dioxide and carbon monoxide evolve at relatively lower temperatures while methane and hydrogen form at higher temperatures. A simplified single-step power law kinetic model is used to fit the experimental data for the gas evolution in fixed bed reactor experiments and the respective kinetic parameters are estimated. The model thus developed is used successfully to predict the flow rate and the composition of gas leaving the pyrolysis zone of gasifier, showing the utility of this approach to obtain an insight into the gasifier operation.

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1. Introduction

The renewable energy derived from biomass not only reduces dependence on fossil fuels but also eliminates the release of excess carbon dioxide to the atmosphere [1]. It also helps in managing waste generated from agriculture, forests, gardens, food processing industries, paper, and pulp industries, and aquatic areas, etc. Garden waste forms almost 20% of the municipal solid waste in metropolitan cities in India [2]. The biomass present in the waste can be converted to solid, liquid, or gaseous fuels to harness its energy potential. It comprises of cellulose (25–50 wt %), hemicellulose (15–40%), lignin (10–40%) along with some amount of extractives and inorganic content [3,4].

There are different thermochemical and biochemical methods to use bioenergy [5]. Pyrolysis is a fundamental thermochemical process that transforms raw biomass directly into gaseous and liquid fuels. The literature is replete on the studies in pyrolysis of woody and non-woody biomass species in the form of agricultural waste, forest residue, marine waste, etc. It is summarized in Table 1.

Various aspects such as analysis of the volatiles produced, effect of pyrolysis conditions, and the role of minerals during thermal heating of biomass, along with kinetic modeling are covered. As can be seen, a very limited research is performed on garden or agro-waste owing to the complexities involved. In most of the cases, it is collected mixed with other wastes (twigs, stems, stumps, branches, leaves, grass clippings, etc.), and its composition varies with seasons, and the age of the tree or type of the crop [6]. This waste has a wide range of ash content from 3 to 20% (dry basis).

While pyrolysis as a standalone process has its own application, it is also an important step in another thermochemical route called gasification wherein, biomass is converted to producer gas comprising of carbon monoxide and hydrogen. There are various types of gasifier designs available depending on the direction of air and gas flow through the biomass bed, to carry out the gasification process [7]. A novel down-draft gasifier (see Fig. 1a) specifically developed for thermal applications (e.g. community cooking), in our research group runs on high-ash garden waste pellets [8]. It contains a slowly moving bed of biomass pellets, and air is sent midway along the height of the gasifier. Biomass pellets are ignited with torch ignition, which results in a development of the combustion zone around the port where air is introduced. The heat is transferred above and below the combustion zone via convection,

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Table 1
Summary of the literature on pyrolysis of lignocellulosic biomass.

Reference	Research work	TGA	FBR	Comments/Conclusion
	Biomass type			
[9]	Switchgrass Wheat straw Eastern red cedar Dry distilled grins	TGA- MS- FTIR.	x	The release of gases is dependent on lignin, cellulose and hemicellulose content.
[10]	Hinoki saw dust, Rice straw, and husks, Larch bark, Corn stalk, Bagasse Palm oil fiber	✓	x	Weight loss profile were compared to a mixture of lignin and cellulose
[11]	Sugarcane bagasse	✓	x	Effect of temperature and heating rate gas formation was studied.
[12]	19 types of grasses	TGA-MS	x	Role of inorganics was observed.
[13]	Wheat straw	✓	x	Various pyrolysis conditions were studied.
[14]	Saw dust	✓	✓	Gases (CO, H ₂), tar formed were analysed. Kinetic parameters for gases were estimated.
[15]	Oriental tobacco, bright tobacco, Burley tobacco, xylan, D-glucose, chlorogenic acid, wheat straw	TG-FTIR	x	Detailed analysis of gases produced was done.
[16]	Sugar cane bagasse, corn cobs, corn stover Eucalyptus Grandis	✓	✓	Bio-oil characterize Role of inorganics was studied.
[17]	Corn stalk	x	Auger Reactor	Detailed characterization of bio-oil.
[18]	Radiata pine	✓	✓	Char structure was analysed using SEM, BET and reactivity was measured.
[19]	Wood, Hazelnut shells, Olive husks, Corn-cobs, Wheat straw, Lucerne pressed cake	✓	✓	Yield of pyrolysis product at different temperature was observed.
[20]	Pectin	x	✓	Volatiles and the char produced were analysed using GC-MS, FTIR, SEM. Char surface area, H/C, O/C ratio varied with temperature
Present study	Dry leaves of Jackfruit, Mango, Eucalyptus and Raintree	✓	✓	Kinetic parameters are estimated for gases and tar produced during pyrolysis.

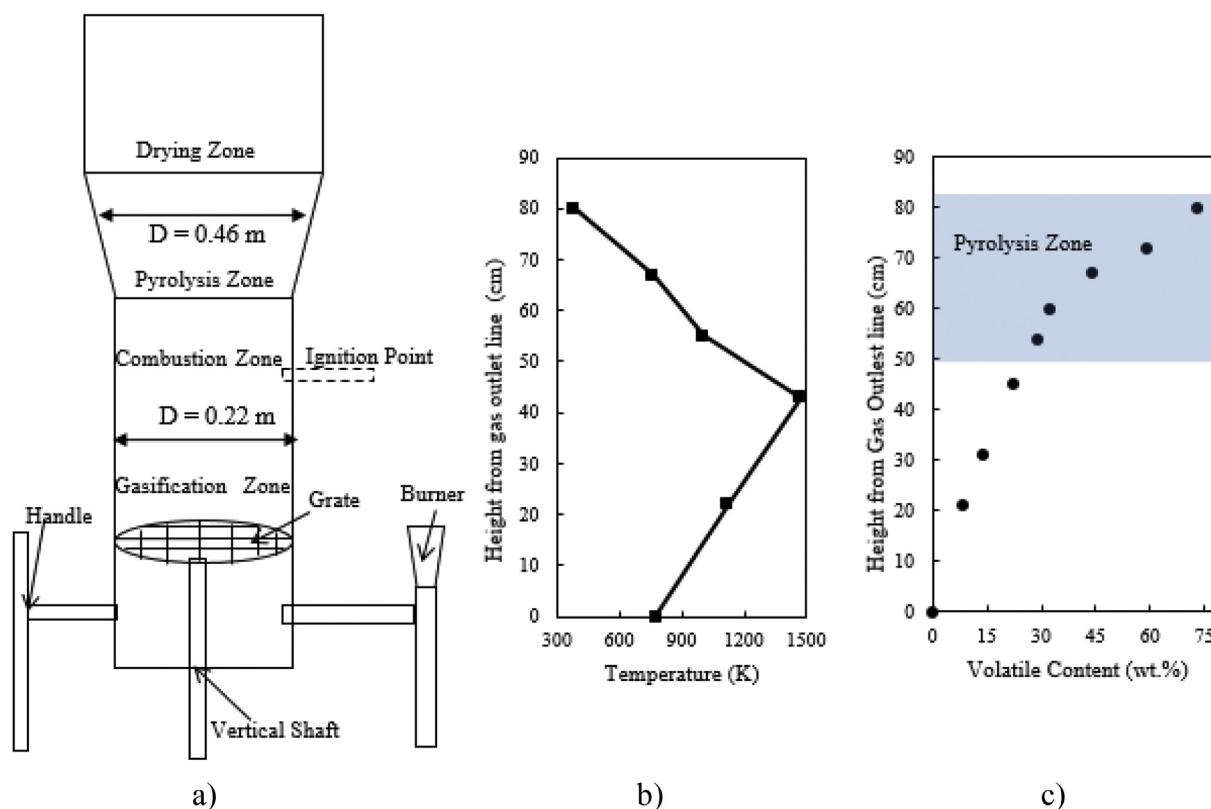


Fig. 1. a) Schematic of a down draft gasifier, b) Temperature profile, and c) volatile content profile along the height.

conduction, and radiation. This results in a variation in temperature along the height of the gasifier. The pyrolysis, combustion and gasification reactions occur predominantly in different zones depending on the respective temperatures. One of the benefits of downdraft gasifier is the relatively small proportion of the tar in the

outgoing gas, because much of it is cracked in the combustion zone. The reaction kinetics for each of these reactions is important to model the gasifier and understand the behaviour of the reactor for the various feedstocks. One approach towards overall model of reactor is to divide its geometry into different zones or

compartments depending on the prevailing reaction, and model these zones separately.

The present work aims at development of a working kinetic model for pyrolysis of garden waste pellets and the estimation of associated kinetic parameters. It may be noted that the model is developed specifically in context of downdraft gasifier. It can help predict the amount and composition of the gas supplied by the pyrolysis zone to the combustion zone thereby making the overall reactor modelling and simulation exercise a lot easier. It may be noted that the intra-pellet transport limitations interfere with the intrinsic reaction kinetics, and that the model developed here is a working kinetic model applicable in the conditions mentioned. Nonetheless, the work presents an approach towards accounting for pyrolysis in reactor model, which can be extended easily to different biomass feeds in downdraft gasifier.

2. Materials and method

The lignocellulosic garden waste was collected from roadside, IIT-B, Mumbai campus. The leaves collected from different trees were washed thoroughly to remove the upper layer of soil and then dried in an oven. These dried leaves were milled and sieved to obtain a particle size in the range of 90–150 μm as the further reduction in size is difficult by normal milling practices. The feedstock was characterized in terms of proximate analysis, ultimate analysis, and chemical composition, i.e., cellulose, hemicellulose, and lignin [3]. The results are given in Table 2. The higher heating values of biomass species were determined using Bomb Calorimeter (IKA C-200) and they were found to be in the range 16–22 kJ/mol, which is greater than that of wood. The characterization of the lignocellulosic biomass confirms its potential as a source of energy. The volatile content of this feedstock is in the range of 65–74% that makes it more promising for pyrolysis study [3].

2.1. Preliminary experiments in downdraft gasifier

The operating windows for temperature and heating rate that prevail in the gasifier were determined by conducting selected experiments. The temperatures at different locations were measured and recorded during its operation. A running gasifier

with 40 kg pellets and air flowing at a flow rate of 23 kg/h, was quenched in the middle of its operation, and samples at different locations were removed to estimate the extent of pyrolysis along the gasifier height. The amount of volatile content was observed to be varying along the height of the gasifier as shown in Fig. 1c. The variation in volatile content shows that pyrolysis occurs throughout the gasifier at varying temperatures. However, it can be seen that almost 60–65% of pyrolysis takes place above the combustion zone. This zone is a source of combustible gases to the combustion zone, and that the composition and flow rate of these gases have a great influence on the temperature of the combustion zone and hence, on the performance of the gasifier. Its realistic prediction is important and the present work is aimed at serving this purpose.

The temperature was measured in the gasifier during the experiment and variations were observed along the length [8]. The rate of change in the surrounding gas temperature relative to pellets can be calculated based on the temperature profile and the pellet residence time. A sample calculation is shown in Appendix - A. The temperature variation rate varies over the range 17–24 K/min in the pyrolysis zone. We have chosen heating rate (20 K/min) in this range, while generating the data in the laboratory fixed bed reactor.

2.2. Kinetic studies in fixed bed reactor

To determine the composition of volatiles containing tar and pyrolysis gases such as CO_2 , CO , CH_4 , and H_2 , a fixed bed reactor setup was used, which could accommodate relatively large amount of biomass (see Fig. 2). Furthermore, it is geometrically similar to the downdraft gasifier. The biomass in the pellet form was used for conducting experiments in the fixed bed reactor. To prepare pellets, garden waste was first shredded and then manually fed to the pelletizer to produce pellets of 15 mm diameter and 50 mm length. The detailed process for pelletization and different steps required to prepare pellets are discussed in our earlier work [21]. The fixed bed reactor setup comprises of a stainless steel (SS) reactor of outer diameter 5 cm and length 30 cm. The reaction can be performed at different heating rates (5–30 K/min) using a programmable electrical furnace. The reactor environment was maintained in a nitrogen atmosphere, at a flow rate of 500 ml/min throughout the experiment. The volume and amount of resultant gases and tar formed, respectively, were determined with respect to time and temperature. The reactor temperature was measured using thermocouple placed at the centre of the reactor. A two-stage ice jacketed condenser was used to separate the tar and the gases produced in the reactor. An acetone impinger was used at the exit of the reactor to efficiently collect the tar produced. The biomass pellets weighing ~60 g for different species were placed in the fixed bed reactor and heated at a rate of 20 K/min, and the pyrolysis gases produced were analysed using GC-TCD. The condensable volatiles were collected in an ice jacketed tar collector. The identified non-condensable gases were CO_2 , CH_4 , CO , and H_2 . The effects of temperature and time on tar formation were studied. The light hydrocarbons were collected in bladders and analysed using GC-TCD, and the residue char was further used for gasification studies. The tar collected was characterized in terms of ultimate analysis as shown in Table 1. The calorific values of both tar and biomass fuel were estimated using the equations given by Demirbas (2017) [22], and, Channiwalla and Parikh (2002) [23], and also by a bomb calorimeter. The HHV values estimated by the two methods were found to be close with the difference of less than 10%. As explained later, the data for gas evolution obtained from the above experiments was fitted using the Arrhenius model (Single Reaction Model) to estimate the kinetic parameters.

It may be noted here that one can always generate intrinsic

Table 2
Characterization of lignocellulosic garden waste (modified from Gupta et al., 2018) [3]).

	Jackfruit	Mango	Eucalyptus	Raintree
Volatile (wt. %)	63	67	76	75
Fixed Carbon (wt. %)	18	23	20	20
Ash (wt. %)	19	10	4	5
C (wt. %)	37	45	52	51
H (wt. %)	5	6	7	8
N (wt. %)	0.7	0.6	1.3	2.6
S (wt. %)	<.5	<.5	<.5	<.5
O ^a (wt. %)	38	38	35	33
Cellulose (wt. %)	25 $\pm$ 5	40 $\pm$ 1	37 $\pm$ 2	28 $\pm$ 2
Lignin (wt. %)	19 $\pm$ 4	29 $\pm$ 1	23 $\pm$ 3	38 $\pm$ 2
Hemicellulose (wt. %)	3 $\pm$ 1	11 $\pm$ 1	14 $\pm$ 1	19 $\pm$ 1
HHV (MJ/kg)	16	18	20	22
Ultimate analysis of Tar				
C (wt. %)	70.3	69.8	75.2	73.1
H (wt. %)	8.4	9.1	10.6	9.7
N (wt. %)	2.3	1.6	1.2	4.9
S (wt. %)	<0.5	<0.5	<0.5	<0.5
O ^a (wt. %)	18.2	18.5	12.4	11.7
Ash content of Tar (wt. %)	0.8	1	0.6	0.62

^a Calculated by difference.

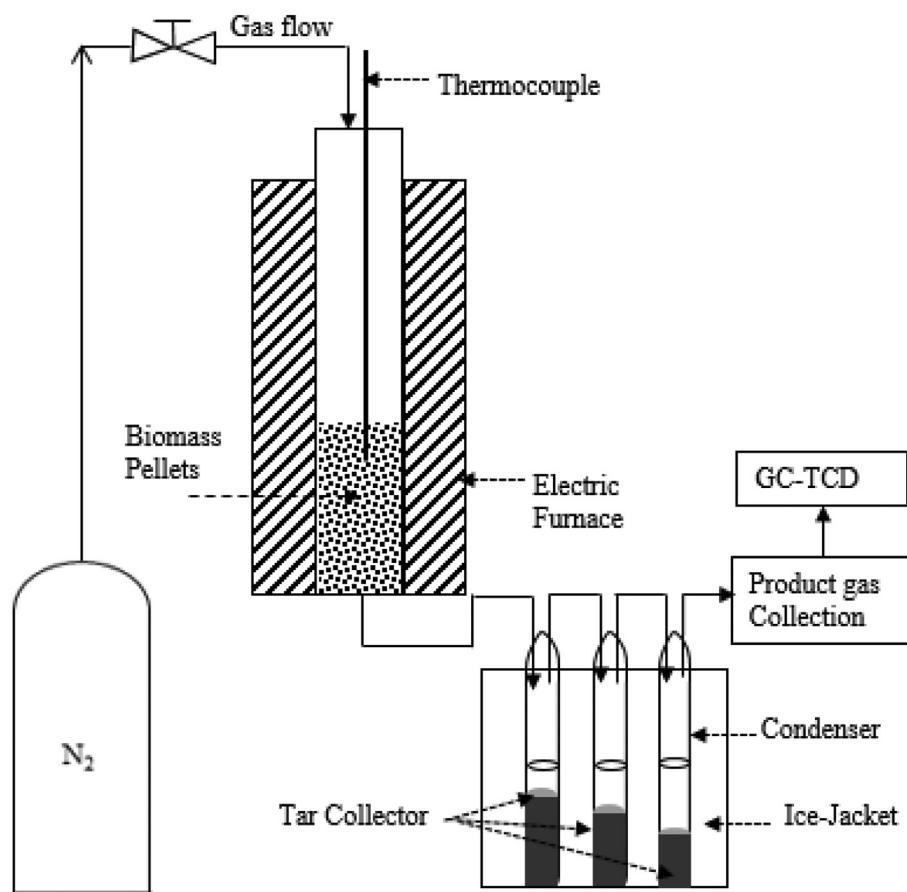


Fig. 2. Schematic of fixed bed reactor setup.

kinetics using Thermogravimetric analyser. However, since the amount of biomass used in TGA is very small, the gases evolved cannot be measured accurately. Furthermore, the TGA conditions and gasifier conditions are quite different and as explained later the intrinsic kinetics may be affected by internal and external transport limitation in the gasifier. Hence, instead of measuring the intrinsic kinetics, our objective is to determine a lumped kinetics that would work well under the conditions that prevail in the gasifier. To check the effect of transport limitations, experiments were conducted with powdered sample and a single pellet. The powdered sample of 5 mg and a single pellet of 2.2 g were used in the TGA instruments. The sample was heated at a heating rate of 10 K/min. It is the maximum heating rate that can be used in the instrument, as at higher than this heating rate, the volatiles produced block the vent line of the instrument. For further comparison of these results (TGA: powder and pellet) with that in the Fixed Bed Reactor, the experiments were conducted in a Fixed Bed Reactor with a heating rate of 10 K/min.

3. Results and discussion

3.1. Gas evolution in FBR

Fig. 3 shows the gas evolution profile with a change in temperature for biomass pyrolysis at 20 K/min. It is clear that the formation of CO_2 and CO starts early in the temperature range of 600–800 K. CO_2 evolution is related to cracking of functional

groups such as $\text{C}=\text{O}$ (carbonyl), and COOH [24–26]. CO gas is evolved due to cracking of groups like $\text{C}-\text{O}-\text{C}$ (carboxyl) and $\text{C}=\text{O}$ (carbonyl) present majorly in cellulose and hemicellulose [26]. The two peaks for CO formation are observed in the temperature range of 650–800 K and above 900 K. The first peak is associated with the decomposition of $\text{C}-\text{C}$ and $\text{C}-\text{O}$ bonds originated from cellulose and hemicellulose, while the peak at higher temperatures is due to the decomposition of lignin. The formation of methane is associated with cracking of methoxy group ($\text{O}-\text{CH}_3$), which is majorly present in aromatic part of lignin [24,25]. The methane formation is directly associated with the lignin content in biomass. The hydrogen formation starts at a relatively high temperature (>700 K), with the maximum formation in the temperature range of 930–1000 K. The formation of hydrogen is associated with cracking of $\text{C}-\text{H}$, and $\text{C}=\text{C}$ bonds present in aromatic groups and is also due to cracking of heavy volatiles at high temperatures [27,28]. A detailed description on the breakage of several organic compounds present in biomass is discussed in Appendix B. The experiments were conducted until hydrogen gas formation ceases. The amount of tar produced during the pyrolysis process at 20 K/min was in the range of 190–260 g/m³ of the total gas produced for these biomasses. The calorific value of tar was estimated using the data on ultimate analysis (Channiwalla and Parikh (2002) [23]), and also by bomb calorimeter. The calorific values of the tars produced from all the biomass species were found to be in the range, 31–35 MJ/kg. While the extent of tar formation varied from biomass to biomass, interestingly, the composition of non-

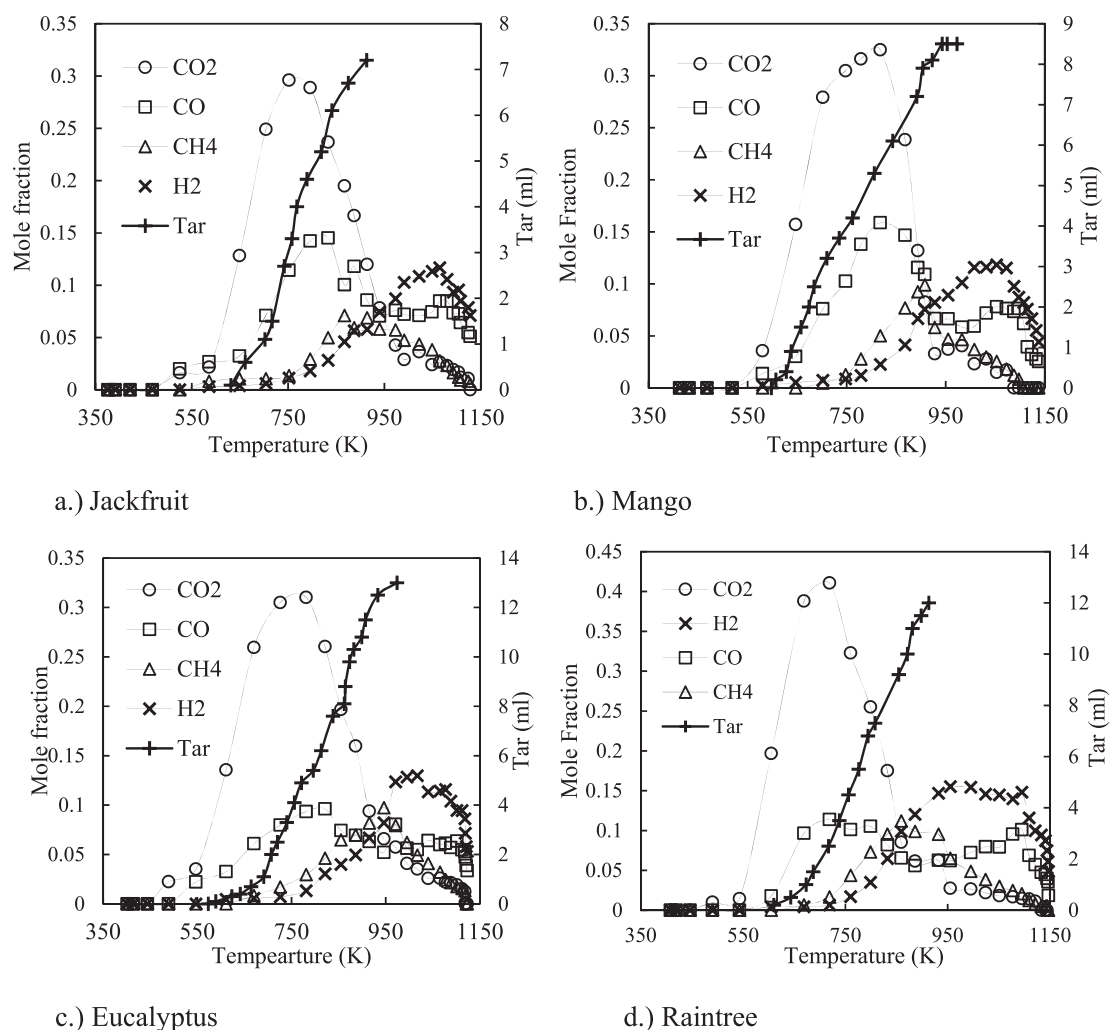


Fig. 3. Gas evolution versus temperature at a heating rate of 20 K/min.

Table 3

The compositions of pyrolysis products for different biomasses at a heating rate of 20 K/min.

Biomass	Gases (wt. %)				Tar (wt. %)	Char (wt. %)
	CO ₂	CO	H ₂	CH ₄		
Jackfruit	20.2	10	~1%	2.3	18	47.5
Mango	21	9.5	~1%	2.4	22	43
Eucalyptus	21	8.3	~1%	2.7	34	31.5
Raintree	22	8.1	~1%	2.7	32	33

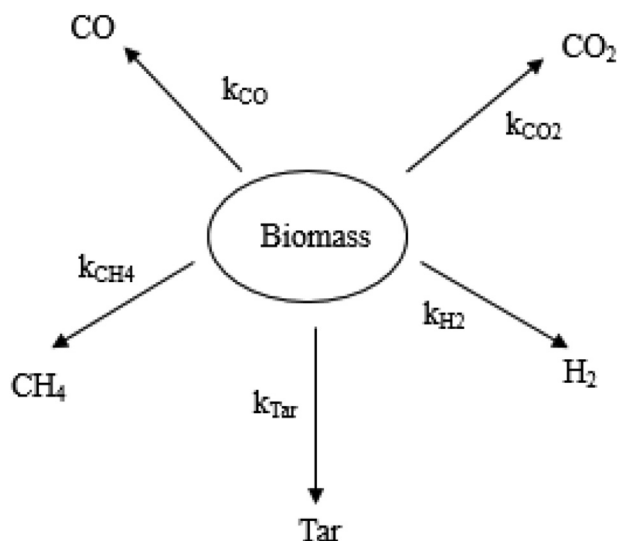
condensable gases was found to be almost similar for all of them. This is in accordance with the literature [27,29].

The amount of non-condensable gases, char and tar produced are tabulated in Table 3. The calculations for the closure of material balance was performed, and the error was found to be less than 2%. As an example, 54 gms of dry biomass yields CO₂: 12 gms, CO: 4.4 gms, H₂: 0.6 gms, CH₄: 1.5 gms, Tar: 17.4 gms, and char 18 gms. While the compositions of pyrolysis gases in each case are more or less similar for all the species, significant differences are observed mainly in the amounts of tar and char produced. Biomass is a complex mixture of several organic and inorganic compounds, and

numerous reactions occur simultaneously during pyrolysis [3]. Therefore, the exact reaction mechanism that results into char and tar formation is not known. Furthermore, the formation of char and tar formation also depends on reactions such as cracking, hydrolysis, aromatization, and repolymerisation/condensation that occur simultaneously throughout the pyrolysis process. These reactions are dependent on heating rate, temperature, cellulose/lignin content and the presence of inorganic materials [30,31]. As seen in Table 3, the amounts of tar produced from eucalyptus and raintree are higher as compared to other two species having higher ash contents. The residue left at the end is considered as char, and the more the ash content the more is the char formation [30,32].

3.2. Kinetics of gas evolution

We propose a simple single reaction model for biomass devolatilization into gases. In this mechanism, the biomass converting to certain gas is considered as a one-step process (Scheme-1). It may be noted that there are more rigorous multi-step models reported in literature to obtain better insight into pyrolysis kinetics. However, as mentioned before, since the objective here is to demonstrate an approach towards reactor modelling, we restrict ourselves to a simplified apparent kinetics which explains the fixed bed



Scheme 1. Different parallel reactions taking place in pyrolysis of biomass. The k_s on the corresponding arrows indicate the respective rate constants.

reactor data reasonably well. We call it as a lumped model because it lumps multiple complex pyrolysis steps and, more importantly the intraparticle and external transport effects, into a single-step power-law kinetics.

The evolution of species during pyrolysis of biomass can be represented in general form as,



Where “ i ” is the gaseous species generated in pyrolysis. Assuming pyrolysis is described by independent, parallel reactions, the reaction rate can be modelled as

$$\frac{dx_i}{dt} = k_i(x_i^* - x_i)^n \quad (2)$$

where, x_i is the accumulated yield of evolved species “ i ” in (ml/g) at time t or temperature T , and x_i^* is the ultimate yield (ml/g) of species “ i ” [33].

The kinetic parameters were determined for the gases evolved and tar formed during the pyrolysis of biomass at a heating rate of 20 K/min using the Single Reaction Model as shown in Equation (3) (solved from Equation (2)).

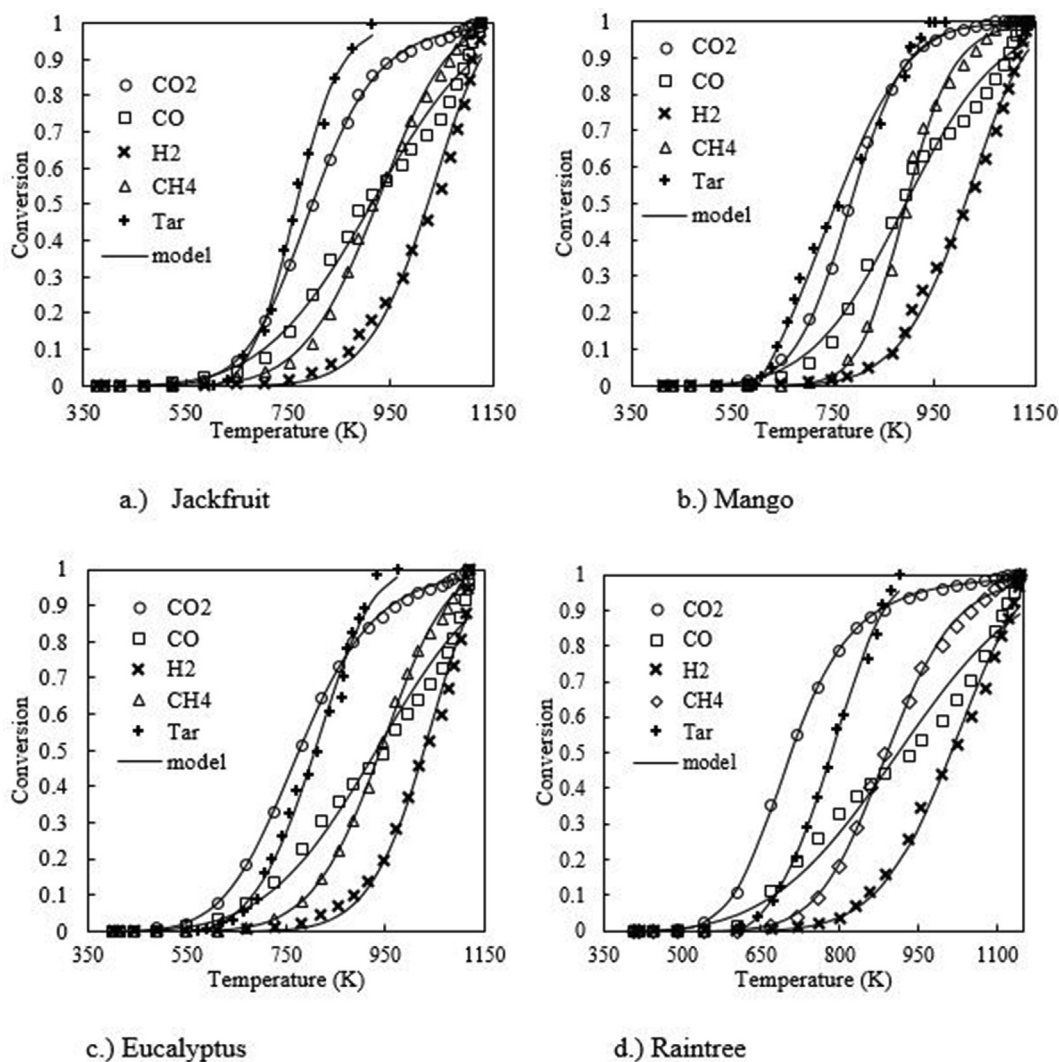


Fig. 4. Gas evolution kinetic model fitting for a heating rate of 20 K/min.

Table 4

Gas evolution kinetics for pyrolysis at 20 K/min.

	Jackfruit			Mango			Eucalyptus			Raintree		
	A (min ⁻¹)	E _a (kJ/mol)	n	A (min ⁻¹)	E _a (kJ/mol)	n	A (min ⁻¹)	E _a (kJ/mol)	n	A (min ⁻¹)	E _a (kJ/mol)	n
CO ₂	2.3E+03	60.7	1.6	8.2E+03	67.5	1.48	213.7	45.3	1.57	1.3E+04	61.5	2.17
CO	9.9E+03	77.2	2.5	36.7	42.7	1.21	2.31	28.2	0.46	5.31	31.1	1.01
CH ₄	450	61.3	1.09	5.4E+06	123	1.83	953	67.9	1.09	5.1E+03	74.2	1.56
H ₂	50.8	55.4	.013	51.7	54.2	0.24	917	77.6	0.29	34	51.1	0.27
Tar	3.9E+04	75.3	1.3	1.21	15.8	0.57	213	48.9	0.89	105	43.1	0.68

$$\frac{dx}{dT} = \frac{A_i}{\beta} \exp(-E_a / RT)(1 - x)^n \quad (3)$$

where, E_a is the activation energy, A is the pre-exponential factor, β is the heating rate (dT/dt), T is temperature and, x is the conversion [34].

$$x = \frac{V_i / (\text{Initial Biomass Weight})}{V_i^* / (\text{Initial Biomass Weight})} = \frac{x_i}{x_i^*} \quad (4)$$

where, " V_i " the amount of volatile produce (ml) at a particular

temperature T , and V_i^* is the maximum amount of volatiles produce (ml).

The experimental data was fitted using non-linear regression in MATLAB. Data fitting for the gas evolution kinetics is shown in Fig. 4. Table 4 gives the estimated values of the kinetic parameters for the gases evolved during the pyrolysis process at a heating rate of 20 K/min for all the biomasses. The mean square error was calculated for all the experimental and theoretical data, and it was observed that CO had higher value of mean square error i.e. 0.001, as compared to other gases. A similar trend in the formation of individual gases has been reported in literature for the pyrolysis of rice husk and babul wood [33,35]. As shown later, this apparent reaction kinetics, with the associated parameters, is further used to predict the yield and product distribution in the pyrolysis zone.

3.3. Check for transport limitations

As mentioned before, the main reason for employing the simplified kinetics is to avoid multiscale and rigorous modelling associated with transport limitations. The heat transfer resistance affects the rate of reaction with the increase in particle size and heating rate. In this section, we provide evidence of such transport limitation by performing experiments in TGA on powdered biomass and a single pellet, separately. The data thus obtained is further compared with the kinetics obtained in our laboratory fixed bed reactor. The results are shown in Fig. 5. The TGA runs (powdered sample) represent intrinsic kinetics. The pyrolysis kinetics for the pellets, both in TGA and that in the fixed bed reactor, were observed to be slower than the intrinsic kinetics. The difference between the rates of reaction on powder and pellet indicated the presence of strong heat transport limitation in case of reactions with pellets. The slight difference between the rates of pellet reactions in TGA and that in the fixed bed reactor (FBR) may be attributed to the external heat transport limitation in FBR. While the TGA results were found to be independent of heating rate, the transport limitations in fixed bed reactor can also be realised by varying the heating rate.

When the experiments were conducted at different heating rates in the fixed bed reactor using raintree pellets, the variations in the yield of char, tar and gases were observed. With the increase in heating rate, char yield decreased, while tar and non-condensable gases increased. When the heating rates of 5, 10, 15, 20, 30 K/min were employed to pyrolyze the raintree pellets, the char produced was 33%, 30%, 29%, 27%, and 25% respectively. In the same way, the amounts of tar increased from 15% to 39% when heating rate was increased from 5 to 30 K/min. At slower heating rate, volatiles produced spent longer time in the reactor, which resulted into secondary reactions. The secondary reactions include further cracking, re-polymerization, re-condensation, that result into char formation and cracking of heavy volatiles [36].

With an increase in the heating rate, the heat transfer from the hot reactor wall to the pellets starts showing up its effect. Owing to

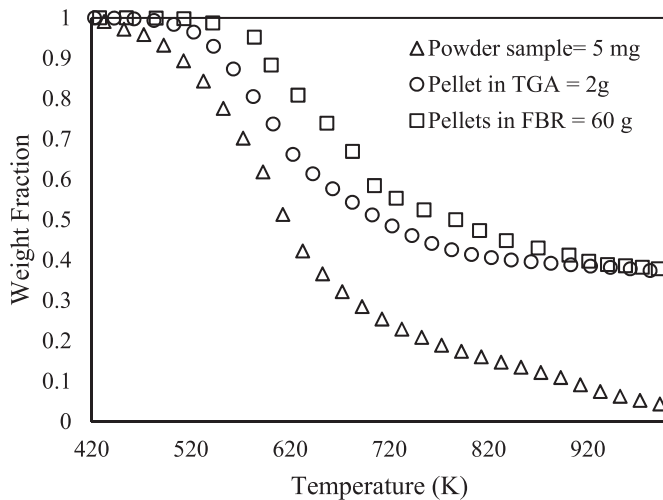


Fig. 5. Weight loss versus Temperature for TGA and FBR.

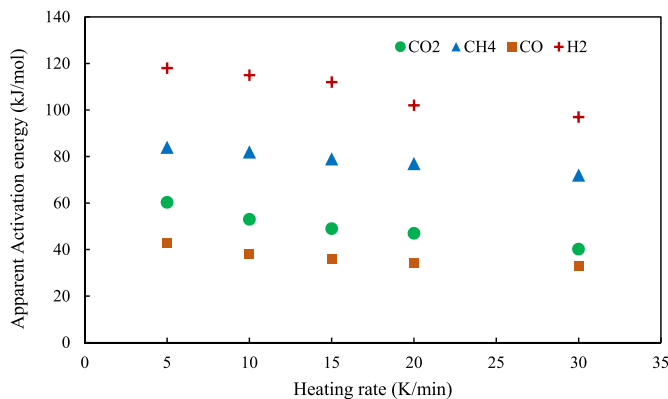


Fig. 6. Variation in apparent activation energy of non-condensable gases with heating rate for pyrolysis of Raintree Pellets.

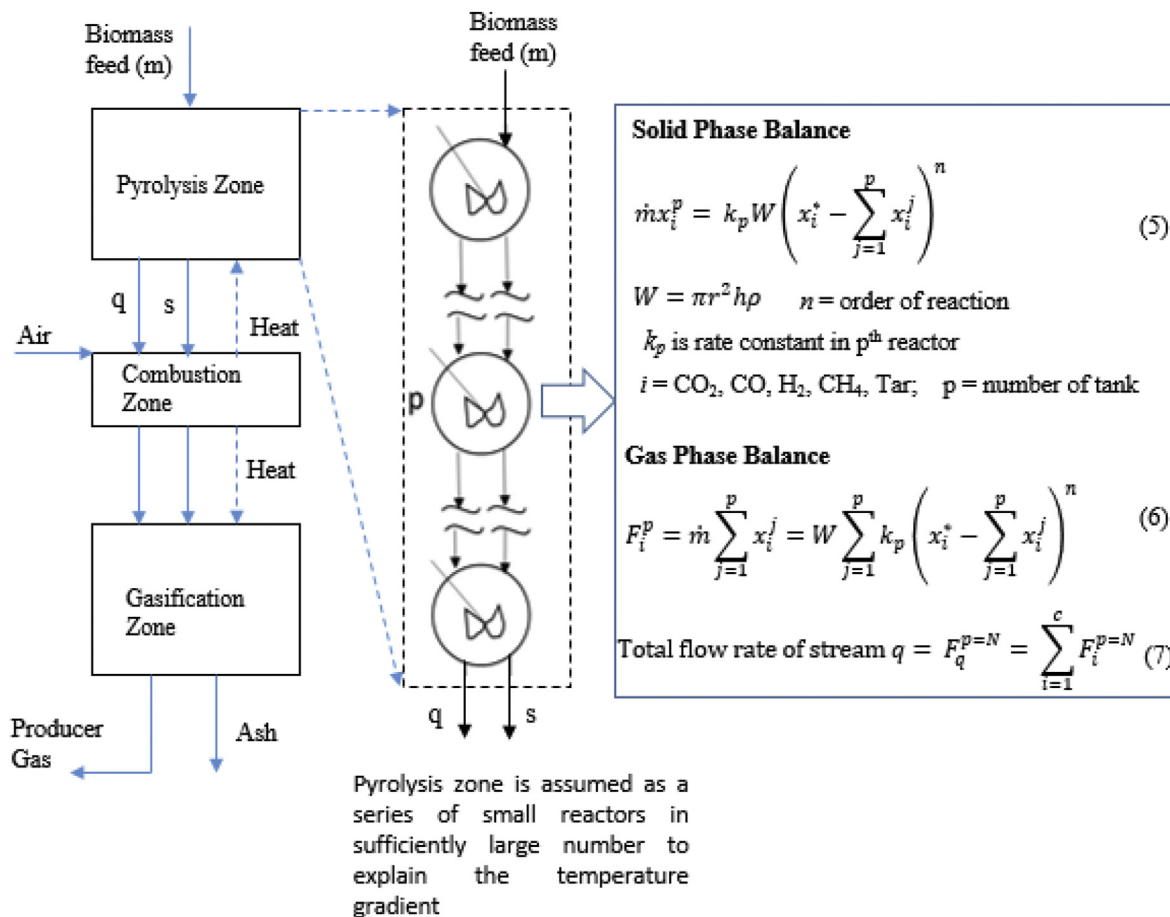


Fig. 7. Compartment model for Downdraft gasifier and Pyrolysis zone in Downdraft gasifier.

these limitations, the rate of weight loss in the fixed bed reactor is less as compared to rate of weight loss in TGA, under otherwise similar conditions as shown in Fig. 5. With the increase in heating rate, the temperature gradient in the pellet also increases. At a higher heating rate, temperature of the wall keeps on increasing, but the actual temperature inside the pellet remains low as compared to external temperature due to transport limitations. This is supported by the variation in apparent activation energy for the gases produced at different heating rates, as shown in Fig. 6. The diffusion and heat transfer effects in the bed as well as within pellets seem to be affecting the reaction rate significantly [37,38]. It is therefore necessary to incorporate the heat transfer limitation and perform kinetic modelling at pellet level if one desires to link the intrinsic kinetics with the fixed bed reactor kinetics. However, it is out of the scope of this work. We intend to demonstrate an approach wherein, an apparent or lumped kinetics can also be used to estimate the performance of the pyrolysis zone.

3.4. Validation of the kinetic model

The working kinetic model developed in the earlier section is used here to explain the results of the reactor runs. Fig. 7 shows the compartment model of the gasifier which is divided in three different zones - pyrolysis, combustion and gasification based on the dominating reaction in the respective zones. The objective is to estimate the flow rate and composition of the stream leaving the pyrolysis zone which acts as a source of mass and heat to the combustion zone. For modelling purpose, the pyrolysis zone is

assumed to be a cascade of back-mixed reactors with co-current flow of biomass and the product gas. The model for every reactor in the cascade is also given in Fig. 7. It involves material balances on both solid and gas sides. The energy balance is not considered here, and the temperature profile is directly adopted from the experiments (see Fig. 1). Hence, it must be noted that the model given in Fig. 7 is not a complete model in a true sense, and it is used here only to demonstrate the applicability of the kinetics developed.

The Equations (5–7) in Fig. 7 are solved using MATLAB nonlinear algebraic equation solver to obtain the profile of volatile content in the pyrolysis zone along the height of the gasifier, and to calculate the flow rate, composition and calorific value of the gas stream (stream q in Fig. 7) that acts as a source to the combustion zone. The pyrolysis zone was divided into sufficiently large number (16) of equal sections, each of height 2 cm, and each section is considered as a completely back-mixed reactor of uniform composition. The volatile content (wt. %) predicted at different locations are plotted in Fig. 8. The amount of residual volatile content in biomass at the exit of pyrolysis zone (at 54 cm length) is 28% (w/w) i.e. ~ 62% of pyrolysis occurs in the pyrolysis zone. This is in agreement with the experimental results. It may be seen here that the profile of the predicted volatile content agrees well with the experimental data. The flow rate of the outgoing stream from the pyrolysis zone is 120 g/min, with a composition of 32% CO_2 , 11% CO , 5% methane, 1% hydrogen and 50% tar. The calorific value of this pyrolysis gas is estimated as 9.59 MJ/m^3 .

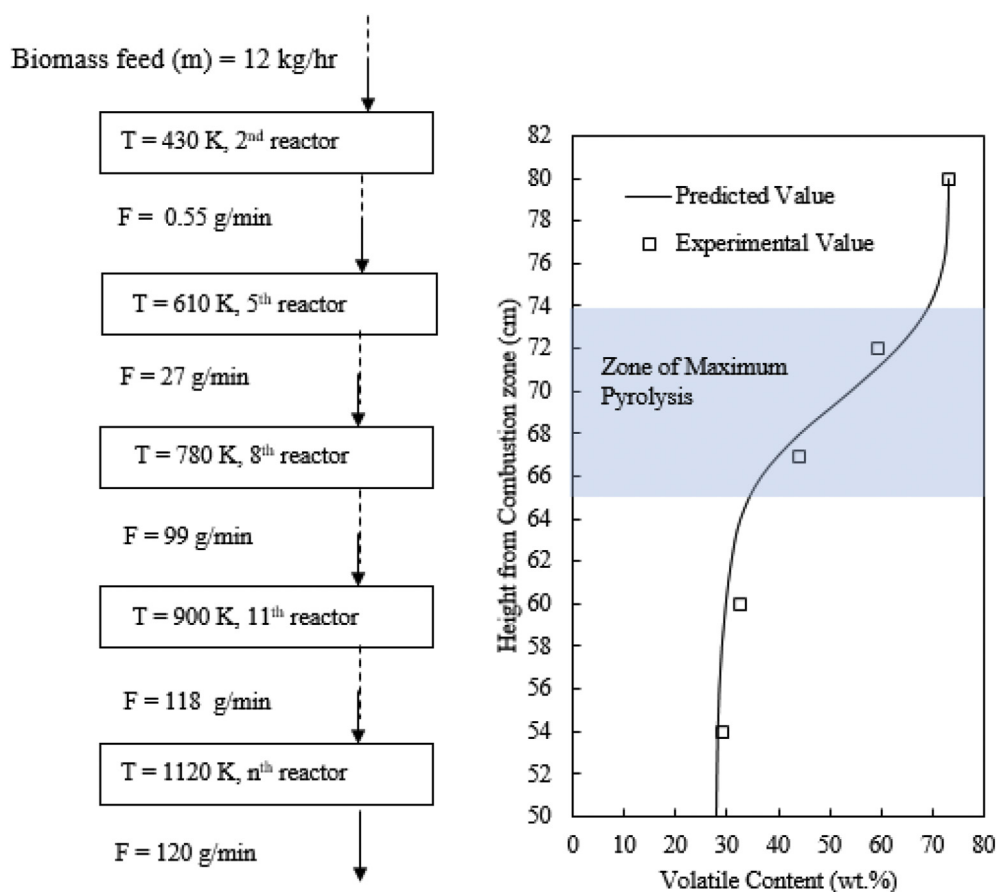


Fig. 8. The comparison of predicted volatile content with the experimental results in Pyrolysis zone of the laboratory downdraft gasifier (air flow rate 23 kg/h) for raintree pellets. Figure also shows the flow rate variation along the height of the gasifier.

4. Conclusion

The pyrolysis of ligno-cellulosic garden waste results in release of CO_2 and CO first due to the decomposition of cellulose and hemicellulose. At temperatures > 673 K, hydrogen and methane are evolved due to the presence of aromatic and methoxy group in lignin. Tar formation was monitored with time and temperature, and the amount of tar produced increases with increase in heating rate. While the compositions of non-condensables are almost similar for all the species, the extents tar produced from eucalyptus and raintree were found to be greater than that from the other two species. The kinetics for the formation of non-condensable gases and tar can be well explained by n th order single-step reaction model that assumes direct conversion of biomass to gases. The values of activation energy for each gas evolution reaction were found to be dependent on the heating rate which indicates that the heat and mass transport limitations at particle level play a crucial role in the kinetics.

Furthermore, the utility of this apparent pyrolysis kinetics is demonstrated to predict the flow rate and the composition of the gas leaving the pyrolysis zone. The predicted results match reasonably well with the experimental findings. This, we believe, is an important step in the development of an exhaustive compartment model of downdraft gasifier.

CRediT authorship contribution statement

Ankita Gupta: Writing - original draft, Investigation, Validation,

Conceptualization, Methodology. **Sanjay Mahajani:** Writing - review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Sample Calculation for heating rate in downdraft gasifier.
 Biomass consumption rate: 12 kg/h.
 Density: 650 kg/m^3
 Radius of gasifier: 0.23 m (upper section), 0.11 m (lower section)
 Distance between two thermocouples: ~15 cm (Height of Gasifier: 67 cm–52 cm)
 Velocity is calculated as,

$$\text{velocity} = \frac{\text{Biomass consumption rate}}{\text{Bulk density} * \text{Area of gasifier}} \quad (\text{A.1})$$

The biomass consumption rate is typically 12 kg/h, the area of the downdraft gasifier used in the present work is 0.038 m² and the density is considered as 650 kg/m³. The linear velocity in the pyrolysis zone thus turns out to be 8 mm/min and the corresponding residence time in a zone of 150 mm is 18.75 min. Now, from the measured temperature profile (Fig. 1), one can calculate the local temperature gradient, dT/dz , where z is the axial distance downward. This gradient when multiplied by the linear velocity calculated above, would give the heating rate for a pellet in terms of K/min, which can be used as a criterion in designing the lab experiments on kinetics using a fixed bed reactor or a thermogravimetric analyser (TGA).

Appendix B

FTIR.

The FTIR analysis of the raintree char prepared at different temperatures is shown in Figure B1. The char was prepared at different temperatures ranging from 473 to 1173 K, and analysed using FTIR to study the change in chemical composition/functional group with increasing temperature. Scanning was carried out in the wavelength range of 400–4000 cm⁻¹ with the step size of 4 cm⁻¹ at a scanning rate of 40. The evidence of bond breakage as represented by a change in IR peak heights was validated by observing gas composition at these temperatures. The change in functional group of char produced on heating biomass at different temperatures would help in explaining the pyrolysis mechanism. The FTIR spectra of raw biomass (dry leaves sample at 303 K) shows the complex transmittance bands, with the various number of functional groups present. At the temperature of 1023 K, biomass gets fully decomposed to char, and there are very little organic functional groups left in the char.

Figure B1 shows that peaks in the range of 3200–3600 cm⁻¹ that correspond to the O–H group [26], and this peak is observed in all the four biomass species. The O–H peak is also associated with glucoside linkages of cellulose, or hydroxyphenyl, guaiacyl, and

syringyl group of lignin [39]. The decomposition of cellulose and lignin is related to breakage of O–H bond. The bands in the range of 2900–3000 cm⁻¹ are usually associated with the aliphatic C–H groups [40]. In all the biomass species, it was observed that on heating, C–H bond peak disappeared after 700 K. This disappearance is associated with the breaking of alkyl groups present, and it results in the formation of methane, ethane and ethylene [41]. In raintree, eucalyptus, and mango, there is a band in the region around 1730 cm⁻¹, which indicates the presence of C=O (aldehyde/ketone) which are related to a free carbonyl compound in the hemicellulose [42]. All the four biomass species have peaks near 1629 cm⁻¹ which is associated with C=C of aromatic compounds as mentioned in Yang et al. (2007) [26]. This peak disappeared in the char produced at a temperature greater than 1000 K. The gas release with an increase in temperature results into the disappearance of various functional groups. The peak in the range 1470–1430 cm⁻¹ is related to the presence of methoxy group (O–CH₃), which results in the release of methane gas at a higher temperature (>700 K) [26]. The peak at around ~1158 cm⁻¹ was found for eucalyptus biomass and is related to the C–O–C bond which shows the presence of cellulose and hemicellulose [43]. The peak in the range ~1053 cm⁻¹ was observed in raintree and eucalyptus at 303 K, which is associated with the presence of C=O, C–3 and C–C bonds which are attributed to cellulose content [42]. The decomposition of cellulose was also seen after 800 K because the peak got diminished after heating biomass at > 800 K. The FTIR peak in the range of 700–900 cm⁻¹ corresponds to the C–H group of aromatic compounds [26] and was found in all the four biomass species. This peak was not observed in the FTIR spectra at room temperature i.e. at 303 K. This peaks were observed on heating biomass char from 523 K to 653 K but later started to diminish after 653 K in all the cases. This appearance of the peak in a particular temperature range is associated with increasing aromaticity which gets broken down on further heating, is confirmed by the disappearance of peak at a higher temperature (>1100 K). Reduction of these bands at higher temperature shows the carbonization due to cracking and rebuilding of aromatic rings, which result in the release of hydrogen gas [41].

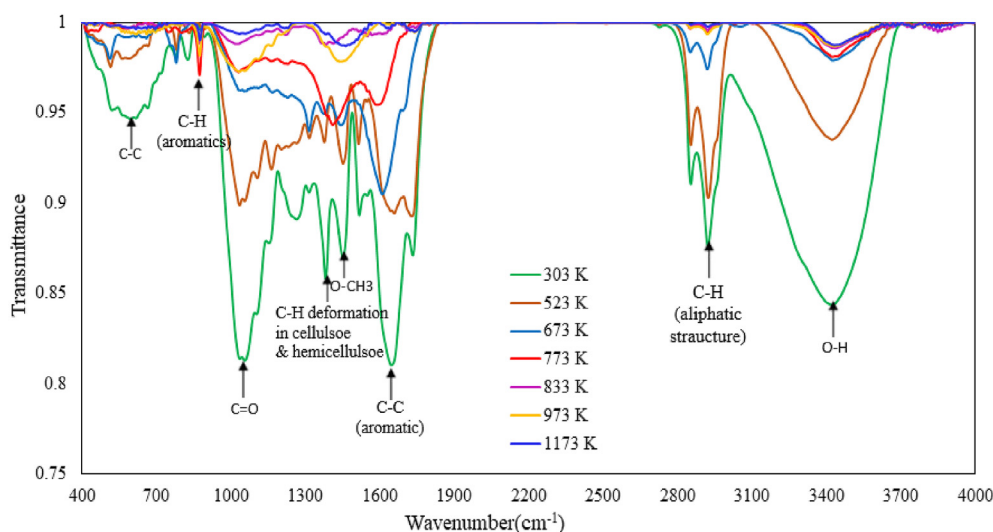


Fig. B1. FTIR analysis of biomass char formed at different temperatures for Raintree Biomass.

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